

A medieval knight with a long beard and hair, wearing ornate armor with fur trim, stands in a dimly lit room. He is looking into a large, ornate mirror. In the reflection, a dragon with glowing red eyes and sharp teeth is visible. The room has a stone wall, a small arched window, and several lit candles in holders. A semi-transparent white box is overlaid on the center of the image, containing the title and author information.

# Duality

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# Motivation

Duality lets us associate to any constrained optimization problem a concave maximization problem, whose solutions lower bound the optimal value of the original problem. What is interesting is that there are cases, when one can solve the primal problem by first solving the dual one. Now, consider a general constrained optimization problem:

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As a consequence:

$$\max_{y \in \Omega} g(y) \leq \min_{x \in S} f(x)$$

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And the Lagrangian, associated with this problem:

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x) = f_0(x) + \lambda^\top f(x) + \nu^\top h(x)$$

## Dual function

We assume  $\mathcal{D} = \bigcap_{i=0}^m \text{dom } f_i \cap \bigcap_{i=1}^p \text{dom } h_i$  is nonempty. We define the Lagrange dual function (or just dual function)  $g : \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$  as the minimum value of the Lagrangian over  $x$ : for  $\lambda \in \mathbb{R}^m, \nu \in \mathbb{R}^p$

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When the Lagrangian is unbounded below in  $x$ , the dual function takes on the value  $-\infty$ . Since the dual function is the pointwise infimum of a family of affine functions of  $(\lambda, \nu)$ , it is concave, even when the original problem is not convex.

## Dual function as a lower bound

Let us show, that the dual function yields lower bounds on the optimal value  $p^*$  of the original problem for any  $\lambda \succeq 0, \nu$ . Suppose some  $\hat{x}$  is a feasible point for the original problem, i.e.,  $f_i(\hat{x}) \leq 0$  and  $h_i(\hat{x}) = 0, \lambda \succeq 0$ . Then we have:

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The term “dual feasible”, to describe a pair  $(\lambda, \nu)$  with  $\lambda \succeq 0$  and  $g(\lambda, \nu) > -\infty$ , now makes sense. It means, as the name implies, that  $(\lambda, \nu)$  is feasible for the dual problem. We refer to  $(\lambda^*, \nu^*)$  as dual optimal or optimal Lagrange multipliers if they are optimal for the above problem.

# Summary

|             | Primal                                                                                                             | Dual                                                                                                           |
|-------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Function    | $f_0(x)$                                                                                                           | $g(\lambda, \nu) = \min_{x \in \mathcal{D}} L(x, \lambda, \nu)$                                                |
| Variables   | $x \in S \subseteq \mathbb{R}^n$                                                                                   | $\lambda \in \mathbb{R}_+^m, \nu \in \mathbb{R}^p$                                                             |
| Constraints | $f_i(x) \leq 0, i = 1, \dots, m$<br>$h_i(x) = 0, i = 1, \dots, p$                                                  | $\lambda_i \geq 0, \forall i \in \overline{1, m}$                                                              |
| Problem     | $f_0(x) \rightarrow \min_{x \in \mathbb{R}^n}$ s.t. $f_i(x) \leq 0, i = 1, \dots, m$ $h_i(x) = 0, i = 1, \dots, p$ | $g(\lambda, \nu) \rightarrow \max_{\lambda \in \mathbb{R}_+^m, \nu \in \mathbb{R}^p}$ s.t. $\lambda \succeq 0$ |
| Optimal     | $x^*$ if feasible,<br>$p^* = f_0(x^*)$                                                                             | $\lambda^*, \nu^*$ if max is achieved,<br>$d^* = g(\lambda^*, \nu^*)$                                          |

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$$-(1/4)\nu^T A A^T \nu - b^T \nu \leq \inf\{x^T x \mid Ax = b\}.$$

Which is a simple non-trivial lower bound without any problem solving.

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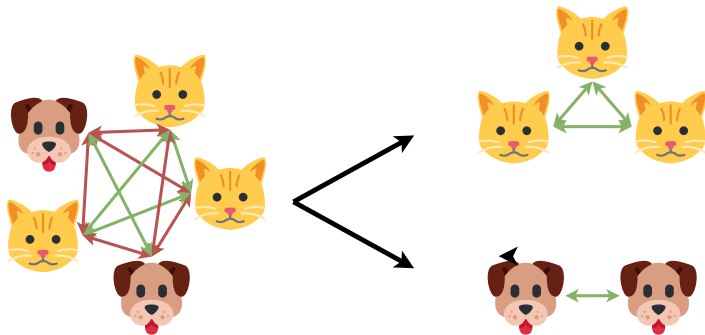


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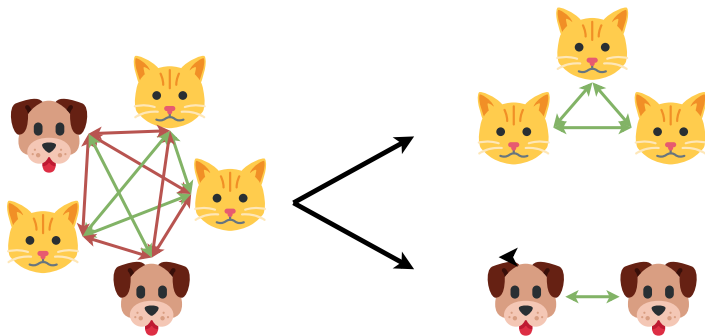


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The coefficient  $W_{ij}$  in the matrix represents the expense associated with placing elements  $i$  and  $j$  in the same partition, while  $-W_{ij}$  signifies the cost of segregating them. The objective encapsulates the aggregate cost across all pairs of elements, and the challenge posed by problem is to find the partition that minimizes the total cost.

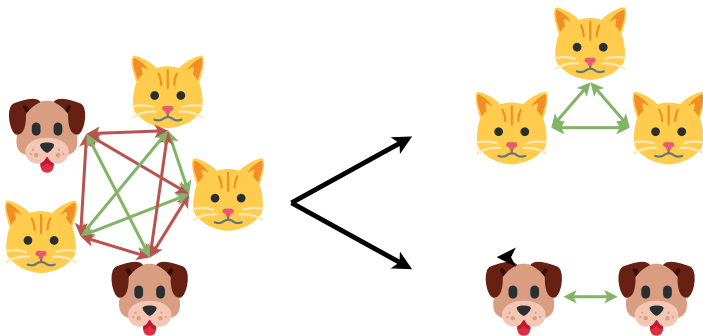


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which is dual feasible, since  $W + \text{diag}(\nu) = W - \lambda_{\min}(W) I \succeq 0$ .

This renders a simple bound on the optimal value  $p^*$ :  $p^* \geq -\mathbf{1}^T \nu = n \lambda_{\min}(W)$ .

## Example. Two-way partitioning problem

We now derive the dual function for this problem. The Lagrangian is expressed as

$$L(x, \nu) = x^T W x + \sum_{i=1}^n \nu_i (x_i^2 - 1) = x^T (W + \text{diag}(\nu)) x - \mathbf{1}^T \nu.$$

By minimizing over  $x$ , we procure the Lagrange dual function:

$$g(\nu) = \inf_x x^T (W + \text{diag}(\nu)) x - \mathbf{1}^T \nu = \begin{cases} -\mathbf{1}^T \nu & \text{if } W + \text{diag}(\nu) \succeq 0 \\ -\infty & \text{otherwise,} \end{cases}$$

exploiting the realization that the infimum of a quadratic form is either zero (when the form is positive semidefinite) or  $-\infty$  (when it's not).

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The code for the problem is available here  Open in Colab

## Strong duality



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- Several sufficient conditions known!
- “Easy” necessary and sufficient conditions: unknown.

## Strong duality in linear least squares

### Exercise

In the Least-squares solution of linear equations example above calculate the primal optimum  $p^*$  and the dual optimum  $d^*$  and check whether this problem has strong duality or not.

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# Slater's condition

## Theorem

If for a convex optimization problem (i.e., assuming minimization,  $f_0, f_i$  are convex and  $h_i$  are affine), there exists a point  $x$  such that  $h(x) = 0$  and  $f_i(x) < 0$  (existence of a strictly feasible point), then we have a zero duality gap and KKT conditions become necessary and sufficient.

## An example of convex problem, when Slater's condition does not hold

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The only point in the budget set is:  $x^* = 0$ . However, it is impossible to find a non-negative  $\lambda^* \geq 0$ , such that

$$\nabla f_0(0) + \lambda^* \nabla f_1(0) = 1 + \lambda^* x = 0.$$

## Useful features of duality

- **Construction of lower bound on solution of the primal problem.**

It could be very complicated to solve the initial problem. But if we have the dual problem, we can take an arbitrary  $y \in \Omega$  and substitute it in  $g(y)$  - we'll immediately obtain some lower bound.



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From the inequality  $\max_{y \in \Omega} g(y) \leq \min_{x \in S} f_0(x)$  follows: if  $\min_{x \in S} f_0(x) = -\infty$ , then  $\Omega = \emptyset$  and vice versa.

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- **Dual function is always concave**

As a pointwise minimum of affine functions.

## Sensitivity analysis

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Let us switch from the original optimization problem

$$\begin{aligned} f_0(x) &\rightarrow \min_{x \in \mathbb{R}^n} \\ \text{s.t. } f_i(x) &\leq 0, \quad i = 1, \dots, m \\ h_i(x) &= 0, \quad i = 1, \dots, p \end{aligned} \quad (\text{P})$$

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One can even show, that when  $\text{P}$  is convex optimization problem,  $p^*(u, v)$  is a convex function.

## Sensitivity analysis

Suppose, that strong duality holds for the original problem and suppose, that  $x$  is any feasible point for the perturbed problem:

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And taking the optimal  $x$  for the perturbed problem, we have:

$$p^*(u, v) \geq p^*(0,0) - \lambda^{*T} u - \nu^{*T} v \quad (1)$$

## Sensitivity analysis

In scenarios where strong duality holds, we can draw several insights about the sensitivity of optimal solutions in relation to the Lagrange multipliers. These insights are derived from the inequality expressed in equation above:

- **Impact of Tightening a Constraint (Large  $\lambda_i^*$ ):**

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These interpretations provide a framework for understanding how changes in constraints, reflected through their corresponding Lagrange multipliers, impact the optimal solution in problems where strong duality holds.

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However, if  $f_i(x^*) = 0$ , meaning the constraint is precisely met at the optimum, then the situation is different. The value of the  $i$ -th optimal Lagrange multiplier,  $\lambda_i^*$ , gives us insight into how 'sensitive' or 'active' this constraint is. A small  $\lambda_i^*$  indicates that slight adjustments to the constraint won't significantly affect the optimal value. Conversely, a large  $\lambda_i^*$  implies that even minor changes to the constraint can have a significant impact on the optimal solution.

# Applications

## Solving the primal via the dual

An important consequence of stationarity: under strong duality, given a dual solution  $\lambda^*, \nu^*$ , any primal solution  $x^*$  solves

$$\min_{x \in \mathbb{R}^n} f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{i=1}^p \nu_i^* h_i(x)$$

Often, solutions of this unconstrained problem can be expressed **explicitly**, giving an explicit characterization of primal solutions from dual solutions.

Furthermore, suppose the solution of this problem is unique; then it must be the primal solution  $x^*$ .

This can be very helpful when the dual is easier to solve than the primal.

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# Solving the primal via the dual

For example, consider:

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where each  $f_i(x_i) = \frac{1}{2}c_i x_i^2$  (smooth and strictly convex).

The dual function:

$$\begin{aligned} g(\nu) &= \min_x \sum_{i=1}^n f_i(x_i) + \nu(b - a^T x) \\ &= b\nu + \sum_{i=1}^n \min_{x_i} \{f_i(x_i) - a_i \nu x_i\} \\ &= b\nu - \sum_{i=1}^n f_i^*(a_i \nu), \end{aligned}$$

where each  $f_i^*(y) = \frac{1}{2c_i} y^2$ , called the conjugate of  $f_i$ .

Therefore the dual problem is:

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This gives:

$$x_i^* = \frac{a_i \nu^*}{c_i}.$$

## Mixed strategies for matrix games

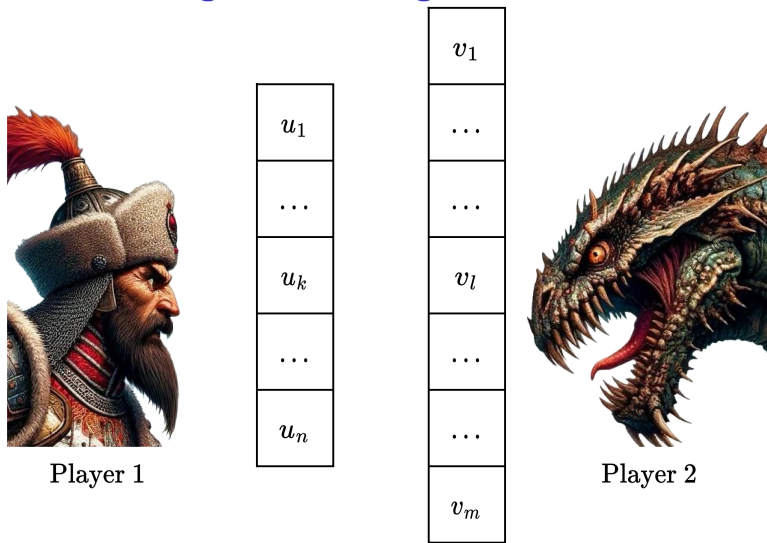


Figure 2: The scheme of a mixed strategy matrix game

## Mixed strategies for matrix games



Player 1

|         |
|---------|
| $u_1$   |
| $\dots$ |
| $u_k$   |
| $\dots$ |
| $u_n$   |

|         |
|---------|
| $v_1$   |
| $\dots$ |
| $\dots$ |
| $v_l$   |
| $\dots$ |
| $\dots$ |
| $v_m$   |



Player 2

In zero-sum matrix games, players 1 and 2 choose actions from sets  $\{1, \dots, n\}$  and  $\{1, \dots, m\}$ , respectively. The outcome is a payment from player 1 to player 2, determined by a payoff matrix  $P \in \mathbb{R}^{n \times m}$ . Each player aims to use mixed strategies, choosing actions according to a probability distribution: player 1 uses probabilities  $u_k$  for each action  $i$ , and player 2 uses  $v_l$ .

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# Mixed strategies for matrix games. Player 1's Perspective



Player 1

|         |
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| $\dots$ |
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Assuming player 2 knows player 1's strategy  $u$ , player 2 will choose  $v$  to maximize  $u^T P v$ . The worst-case expected payoff is thus:

$$\max_{v \geq 0, 1^T v = 1} u^T P v = \max_{i=1, \dots, m} (P^T u)_i$$

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Player 1's optimal strategy minimizes this worst-case payoff, leading to the optimization problem:

$$\begin{aligned} \min \quad & \max_{i=1, \dots, m} (P^T u)_i \\ \text{s.t.} \quad & u \geq 0 \\ & 1^T u = 1 \end{aligned} \tag{3}$$

This forms a convex optimization problem with the optimal value denoted as  $p_1^*$ .

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Conversely, if player 1 knows player 2's strategy  $v$ , the goal is to minimize  $u^T P v$ . This leads to:

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...

...

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...

...

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Player 2 then maximizes this to get the largest guaranteed payoff, solving the optimization problem:

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The optimal value here is  $p_2^*$ .



Player 2

$v_1$

...

...

$v_l$

...

...

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## Duality and Equivalence

It's generally advantageous to know the opponent's strategy, but surprisingly, in mixed strategy matrix games, this advantage disappears. The key lies in duality: the problems above are Lagrange duals. By formulating player 1's problem as a linear program and introducing Lagrange multipliers, we find that the dual problem matches player 2's problem. Due to strong duality in feasible linear programs,  $p_1^* = p_2^*$ , showing no advantage in knowing the opponent's strategy.

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## Conclusion

This formulation shows that the Lagrange dual problem is equivalent to problem Equation 4. Given the feasibility of these linear programs, strong duality holds, meaning the optimal values of Equation 3 and Equation 4 are equal.

## A nonconvex quadratic problem with strong duality

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### Solution

Lagrangian and dual function

$$L(x, \lambda) = x^\top Ax + 2b^\top x + \lambda(x^\top x - 1) = x^\top (A + \lambda I)x + 2b^\top x - \lambda$$

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On rare occasions strong duality obtains for a nonconvex problem. As an important example, we consider the problem of minimizing a nonconvex quadratic function over the unit ball

### Solution

Lagrangian and dual function

$$L(x, \lambda) = x^\top Ax + 2b^\top x + \lambda(x^\top x - 1) = x^\top (A + \lambda I)x + 2b^\top x - \lambda$$

$$\begin{aligned} x^\top Ax + 2b^\top x &\rightarrow \min_{x \in \mathbb{R}^n} \\ \text{s.t. } x^\top x &\leq 1 \end{aligned}$$

$$g(\lambda) = \begin{cases} -b^\top (A + \lambda I)^\dagger b - \lambda & \text{if } A + \lambda I \succeq 0 \\ -\infty, & \text{otherwise} \end{cases}$$

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# References

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- Duality Uses and Correspondences lecture by Ryan Tibshirani course.